

COCOMO Model – Numerical Problems (with Solutions)

Formula (Basic COCOMO)

Effort (E) in person-months (PM)

$$E = a \times (\text{KLOC})^b$$

Development Time (T) in months

$$T = c \times (E)^d$$

Average Staff Size

$$\text{People} = \frac{E}{T}$$

COCOMO Constants

Project Type	a	b	c	d
Organic	2.4	1.05	2.5	0.38
Semi-Detached	3.0	1.12	2.5	0.35
Embedded	3.6	1.20	2.5	0.32

Question 1 (5 Marks)

Estimate effort, development time, and team size for an Organic project of 32 KLOC.

Solution

Effort:

$$E = 2.4 \times (32)^{1.05}$$

$$E = 2.4 \times 38.2 = 91.68 \text{ PM}$$

Time:

$$T = 2.5 \times (91.68)^{0.38}$$

$$T = 2.5 \times 5.56 = 13.9 \text{ months}$$

Team Size:

$$\text{People} = \frac{91.68}{13.9} \approx 7$$

✅ Answer:

Effort = 91.68 PM, Time = 13.9 months, Team = 7 persons

Question 2 (5 Marks)

A Semi-Detached project has a size of 50 KLOC. Calculate effort and development time.

Solution

Effort:

$$E = 3.0 \times (50)^{1.12}$$

$$E = 3.0 \times 79.4 = 238.2 \text{ PM}$$

Time:

$$T = 2.5 \times (238.2)^{0.35}$$

$$T = 2.5 \times 6.78 = 16.95 \text{ months}$$

✓ Answer:

Effort = 238.2 PM, Time = 16.95 months

Question 3 (5 Marks)

An Embedded project requires 120 person-months of effort. Find development time.

Solution

$$T = 2.5 \times (120)^{0.32}$$

$$T = 2.5 \times 5.21 = 13.02 \text{ months}$$

✓ Answer:

Development Time = 13.02 months

Question 4 (5 Marks)

If an Organic project has effort = 60 PM and development time = 12 months, find the average team size.

Solution

$$\text{People} = \frac{60}{12} = 5$$

✓ Answer:

Average Team Size = 5 persons

Question 5 (5 Marks)

Explain why effort increases non-linearly with KLOC in the COCOMO model.

Answer (Theory-based Numerical Insight)

- Effort exponent $b > 1$, hence effort grows faster than size.
- Larger projects need more coordination, testing, and documentation.
- Complexity and communication overhead increase with size.